

Yufan Du

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Education

Peking University

Bachelor of Science in Applied Physics *and* Computer Science
Dual Major

Sep 2021 - Present

Beijing, China

Technical Skills

Proficient in: C/C++, Python

Knowledgeable with: Python ML libraries (PyTorch, NumPy), Linux systems, CUDA/parallel computing, Embedded systems, SQL/Database, JavaScript/HTML, Verilog/VHDL, MATLAB, Tcl, L^AT_EX

EDA tools: Familiar with Cadence Innovus, Cadence Virtuoso.

Tools Used: Git, Vim, Docker, VSCode, CMake/Makefile, Linux utilities

Relevant Scores

Overall GPA: 3.852 (Top 5%)

TOEFL: 104

GRE: 324 (Quantitative + Verbal), 4.0 (Analytical Writing)

Project Experiences

FPGA-Based Snake Game Design Project

- Developed a real-time control mechanism for the Snake game using camera-based hand movement detection.
- Enhanced player interaction by innovative human-computer interface and provided insights for IoT edge device development.

AI-Assisted Schedule Manager

- Created an AI-integrated scheduling application with cross-platform functionality.
- Utilized natural language processing for efficient schedule management and reminders.

Cross-Stage Power Prediction for Integrated Circuits

- Proposed an innovative framework for power prediction, incorporating cross-stage circuit-transformation-aware learning.
- Significantly reduced error rates within shorter time compared to leading commercial tools, as demonstrated on CircuitNet benchmarks.

Relevant Publications

Under Review:

- “PowPrediCT: Cross-Stage Power Prediction with Circuit-Transformation-Aware Learning,” submitted to *Design Automation Conference (DAC24)*.

Relevant Coursework

Academic Year 22-23, Semester 3

Course Name	Credits	Grade	GPA
Introduction to Quantum Information Technology	2	96	3.97
Aural Culture and World Civilization	2	93	3.91

Academic Year 22-23, Semester 2

Course Name	Credits	Grade	GPA
Introduction to Computer Vision	3	95.2	3.96
Anglo-American Tort Law	2	90	3.81
Computer Aided Design of Integrated Circuits	3	95.5	3.96
Practice of Programming in C&C++	3	91	3.85
Introduction to Database Systems	3	92	3.88
Basic Electronics Lab	2	92	3.88
Advanced Design of Digital Circuit and System	4	89	3.77
Modern Physics	3	96	3.97
Theoretical Mechanics (B)	3	92	3.88
Integrated Circuit Devices	4	86.5	3.66

Academic Year 22-23, Semester 1

Course Name	Credits	Grade	GPA
Optics	3	94	3.93
Introduction to Computer Systems	5	97	3.98
Integrated Circuit Manufacturing Technology	3	92	3.88
Analysis and Design of Analog Circuits	5	98	3.99
Methods of Mathematical Physics	3	96	3.97
Fascinating Chemistry	2	91.5	3.87
Music and Mathematics	2	91	3.85

Academic Year 21-22, Semester 3

Course Name	Credits	Grade	GPA
Introduction to Cultural Anthropology	2	93	3.91

Academic Year 21-22, Semester 2

Course Name	Credits	Grade	GPA
Introduction to Seismology	2	97	3.98
Advanced Mathematics A (no. 2)	5	90	3.81
Thermal Physics	3	86	3.63
Electromagnetism	4	99	4
Data Structure and Algorithm (B)	3	95	3.95
Practical English Skills: From Listening to Public Speaking	2	90	3.81

Academic Year 21-22, Semester 1

Course Name	Credits	Grade	GPA
Fitness	1	92	3.88
Principles of Economics	4	89	3.77
Advanced Mathematics A (no. 1)	5	84	3.52
Linear Algebra (B)	4	83	3.46
Mechanics	3	95	3.95
Introduction to Computation (A)	3	96	3.97
An Introduction to Ideological & Moral Culture and Laws	3	91	3.85
Military Theory	2	94	3.93